UE431

Groupe4
(MNEP)

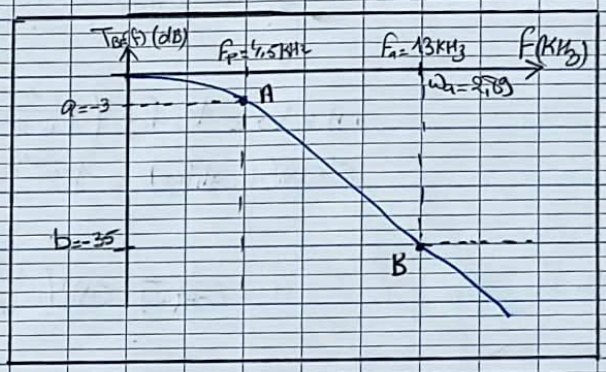
TP1: Synthèse et réalisation de filtres actifs

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A. Preparation:

I. Synthèse LPF Butterworth

$\begin{cases} f_p = 4,5KHz \; ; \; a = -3dB \\ f_a = 13KHz \; ; \; b = -35dB \end{cases}$



. Selectivity:
$K = f_p/f_a = 0,346$
$\omega_a = K^{-1} = 2,89$

. Fonc approx: $K(\omega) = \varepsilon^2 \omega^{2n}$

. $\varepsilon = \sqrt{10^{-a/10} - 1} = 0,997 \approx 1$

gabarit du filter

Proof ↓

. $H(\omega) = \frac{1}{1 + \varepsilon^2 \omega^{2n}}$

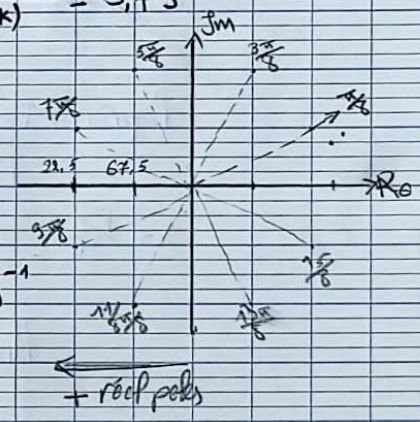
Proof ↓ $n \geq \frac{\ln\left(\frac{10^{-b/10} - 1}{\varepsilon^2}\right)}{2 \ln(1/K)} = 3,fg$

$n_{min} = 4$

$H^2(j\omega) = \frac{1}{1 + \omega^8}$

$1 + \omega^8 = 0 \; : \omega^8 = -1 \Rightarrow \omega = e^{j\frac{(2q+1)}{8}\pi}$

$s = \frac{p}{\omega_p}$



$H(s) = \left[(s - e^{j\frac{5\pi}{8}})(s - e^{j\frac{7\pi}{8}})(s - e^{j\frac{9\pi}{8}})(s - e^{j\frac{11\pi}{8}})\right]^{-1}$

$H(s) = \left[(s^2 + 2\cos(\frac{7\pi}{8})s + 1)(s^2 + 2\cos(\frac{3\pi}{8})s + 1)\right]^{-1}$

$H(s) = \left[(s^2 + 1,847s + 1)(s^2 + 0,765s + 1)\right]^{-1}$

$$H(p) = \frac{1}{\left((\frac{p}{\omega_p})^2 + 1,847\frac{p}{\omega_p} + 1\right)\left((\frac{p}{\omega_p})^2 + 0,765\frac{p}{\omega_p} + 1\right)}$$

proof $\varepsilon \rightarrow$ en A, $\omega = 1$: $a = 10\log\left(\frac{1}{1+\varepsilon^2}\right) \Rightarrow \varepsilon = \sqrt{10^{-a/10} - 1}$

proof $n \rightarrow$ en B: $b \geq 10\log\left(\frac{1}{1+\varepsilon^2\omega_a^{2n}}\right) \Rightarrow 10^{-b/10} \leq 1 + \varepsilon^2\omega_a^{2n} \Rightarrow \omega_a^{2n} \geq \frac{10^{-b/10} - 1}{\varepsilon^2} \Leftrightarrow$

$\Rightarrow 2n\ln(\omega_a) \geq \ln\frac{10^{-b/10} - 1}{\varepsilon^2} \Rightarrow n \geq \frac{\ln\left(\frac{10^{-b/10} - 1}{\varepsilon^2}\right)}{2\ln(\omega_a)}$

II- Synthèse LPF Tchébychev

$$F(\omega) = \frac{1}{1+\varepsilon^2 T_n^2(\omega)} \quad \begin{cases} T_n(\omega) = \cos(n \arccos(\omega)) ; & \omega < 1 \\ T_n(\omega) = \cosh(n \operatorname{arccosh}(\omega)) ; & \omega > 1 \end{cases}$$

- ε :

$-3 = a = 10 \log \frac{1}{1+\varepsilon^2 T_n(\omega)}$ at $\omega = 1$

$$a = 10 \log \frac{1}{1+\varepsilon^2 \cos(n \cos^{-1}(1))}$$

$q \in \mathbb{N}$ $\quad \cos^{-1}(1) = 2q\pi \qquad \cos(n \cdot 2q\pi) = 1 \quad : \quad a = 10 \log \frac{1}{1+\varepsilon^2}$

$$\Rightarrow \varepsilon = \sqrt{10^{-a/10} - 1}$$

- n : $b = 10 \log \left(\frac{1}{1+\varepsilon^2 T_n(\omega_a)} \right) = 10 \log \left(\frac{1}{1+\varepsilon^2 \cosh(n \operatorname{argcosh}(\omega_a))^2} \right)$

$$\varepsilon^2 \cosh(n \cosh^{-1}(\omega_a)) = 10^{-b/10} - 1$$

$$\cosh(n \cosh^{-1}(\omega_a)) = \frac{10^{-b/10} - 1}{\varepsilon^2}$$

$1 + 2m\frac{p}{\omega_0} + \frac{p^2}{\omega_0^2}$

m: damping coef

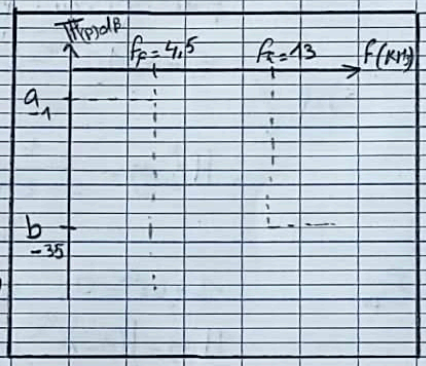
$$n = \frac{1}{\cosh^{-1}(\omega_a)} \cdot \cosh^{-1}\left(\frac{10^{-b/10} - 1}{\varepsilon^2} \right) = n_{min} \qquad \text{argcosh}\left(\sqrt{\frac{10^{-b/10} - 1}{\varepsilon^2}}\right)$$

$$n_{min} = \frac{\cosh^{-1}\left(\sqrt{\frac{10^{-b/10} - 1}{\varepsilon^2}}\right)}{\cosh^{-1}(\omega_a)}$$

$$k = \frac{f_p}{f_a} = 3,46$$

$$\omega_a = 1/k = 2,89$$

$\varepsilon = 0,508$; $n_{min} = (3,13) = 4$



d'après T1 :

$$H(s) = \frac{1}{(1,0136 s^2 + 0,2828 s + 1)(3,5791 s^2 + 2,415 s + 1)}$$

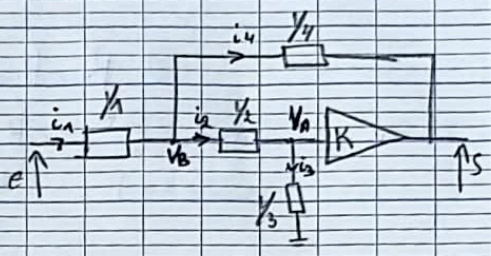
$s = \frac{p}{\omega_p}$

$$H(s) = \frac{1}{(1,0136 s^2 + 0,2828 s + 1)(3,5791 s^2 + 2,415 s + 1)}$$

avec $s = \frac{\omega}{\omega_0}$

III. Réalisation :

1. Sallen-key



H_3 est un montage ampli non inverseur avec op-amp

$i_2 = i_3 \rightarrow (V_B - V_A) Y_2 = Y_3 V_A$ $\quad i_1 = i_2 + i_4$

$V_B = V_A (y_3 + y_2)/y_2$ $\quad y_1 (e - V_B) = Y_2 (V_B - V_A) + y_4 (V_B - V_S)$

$S = K V_A \Rightarrow y_1 \left(e - \frac{V_A}{y_2}(y_3+y_2)\right) = y_2 V_A \left(\frac{y_3+y_2}{y_2} - 1\right) + y_4 \left(V_A \frac{y_2+y_3}{y_2} - V_S\right)$

$$\rightarrow y_1 e - \cancel{K} S y_1 \frac{y_3+y_2}{K y_2} = \frac{y_2}{K} \cancel{K} S \frac{y_3}{y_2} + y_4 \cancel{K} S \left(\frac{y_2(\frac{1}{K}-K)+y_3}{K y_2}\right)$$

$$y_1 e = S\left[\frac{(y_3+y_2)y_1}{K y_2} + \frac{y_2 y_3}{K y_2} + \frac{y_4 y_2 (1-K) + y_3 y_4}{K y_2}\right] = \frac{y_3 + y_2 + y_2 y_3 + y_3 y_4 + y_4 y_2 (1-K)}{K y_2} S$$

$$S/e = \frac{K y_1 y_2}{y_1(y_3+y_2) + y_2 y_3 + y_4 y_2 (1-K) + y_3 y_4} = \frac{K y_1 y_2}{(y_1+y_4)(y_2+y_3) + y_2 y_3 - y_4 y_2 K}$$

$$H(p) = \frac{K \frac{1}{R_1 R_2}}{\left(\frac{1+C_4 R_1 p}{R_1}\right)\left(\frac{1+C_3 p R_3}{R_3}\right) + \frac{C_3 p}{R_2} - \frac{C_4 p}{R_2} K} = \frac{K}{(1+C_4 R_1 p)(1+C_3 p R_3) + R_1 C_3 p - K R_1 C_4 p}$$

$$H(p) = \frac{K}{R_1 R_2 C_3 C_4 p^2 + (C_4 R_1 + C_3 R_3 + C_3 R_1 - K C_4 R_1) p + 1}$$

Stable si $m > 0$

pas seul m loi

$H(p) = \frac{K}{\frac{1}{\omega_0^2} p^2 + \frac{2m}{\omega_0} p + 1}$ $\quad \omega_0 = (R_1 R_2 C_3 C_4)^{-1/2}$, $A = K$, $m = \frac{C_4 R_1 + C_3 R_3 + C_3 R_1 - K C_4 R_1}{2\sqrt{R_1 R_2 C_3 C_4}}$

- pour régler m changer R_1 ; ω_0 reste indépendant

$$m = \frac{C_4 R_1 (1-K) + C_3 (R_1 + R_3)}{2\sqrt{\quad}}$$

- pour régler ω_0 sans changement de m ; il faut rendre m indépendant sur $R_1 - C_4$

par $R_1 = R_2 = R$; $C_1 = C_2 = C$

$$m = \frac{CR(1-K) + 2CR}{2CR} = \frac{1-K}{2} + 2 = \frac{3-K}{2}$$

$m = \frac{3-K}{2}$ indépendant de ω_0 ;

2. Realisation LPF Butterworth

$K = 3 - 2m$ $\quad m = \frac{3-K}{2} \Rightarrow K = 3 - 2m$; $\omega_0 = \frac{1}{RC} \Rightarrow R = \frac{\omega_0}{C}$

$\omega_0 = 2\pi f_a$

$R = 7{,}525\ K\Omega$

$\begin{cases} K_1 = 1{,}152 \\ K_2 = 2{,}27 \end{cases}$

$m = \frac{3-K}{2} \rightarrow K_{lim} = 3 = 1 + \frac{R_2}{R_1} \Rightarrow R_2 \leq 2R_1$

$R_{2lim} = 13{,}6\ K\Omega$

3. Realisation LPF Tchebychev:

$K = 1 + \frac{R_2}{R_1} = 2$; $m = \frac{1}{2} \Rightarrow 2m = 1$ calcul simplifié

$$H(p) = \frac{K}{R_1 R_2 C_3 C_4 p^2 + (C_4 R_1 (1-K) + C_3 (R_1 + R_2)) p + 1}$$

$\hookrightarrow -C_4 R_1 + C_3 R = R(-C_4 + 2C_3) \Rightarrow$ simplif $C_4 = C_3 \Rightarrow CR$

$-CR_1 + C(R_1 + R_2)$

$$H(p) = \frac{2}{R_1^2 C^2 p + CR_1 p + 1}$$

$\begin{cases} 1{,}0136 = R_1^2 C^2 \\ 0{,}5656 = RC \end{cases}$ $\quad \left(\begin{cases} m = \frac{1}{2} \\ \Omega = \frac{1}{RC} \end{cases} \right)$

$$H(p) = \frac{2}{R_1' R_2' C^2 p^2 + CR_2 p + 1}$$

cellule 1 $\begin{cases} 1{,}0136 = R_1 R_2' C^2 \Rightarrow R_1' = \frac{1{,}0136}{C^2 R_2'} \\ 0{,}5656 = CR_2 \Rightarrow R_2' = \frac{0{,}5656}{C} \end{cases}$ $\quad R_1' = 763{,}025\ M\Omega$ $\quad R_2' = 60{,}17\ M\Omega$

cellule 2 $\begin{cases} 3{,}5191 = R_1 R_2 C^2 \\ 2{,}41 = CR_2 \end{cases} \Rightarrow$ $\quad R_1' = 315{,}97 \times 10^6\ \Omega$ $\quad R_2' = 512{,}76 \times 10^6\ \Omega$

$m = \frac{CR_2}{2C\sqrt{R_1 R_2}}$ $\quad m_1 = 0{,}14$ $\quad m_2 = 0{,}63$

m_1 est presque en instabilité

3- Realisation d'un PF Tchebychev:

$p = \frac{\Omega}{\omega_0}$

$H(p) = \frac{K}{\frac{R_1 R_2 C^2}{a_1} p^2 + \frac{C(R_2 + R_1(2-K))}{a_2} p + 1}$ pour le p'en donner plus simple CR_2 avec $K_2 = 2$

$R_2 = \frac{a_2}{\omega_0 C}$; $R_1 = \frac{a_1}{\omega_0^2 R_2 C^2}$

$R_1 R_2 C^2 \frac{\omega^2}{\omega_0^2}$

$a_2 = R_1 R_2 C^2 / \omega^2$

$\frac{a_1}{\omega_0} = R_1 R_2 C^2$

Cellule 1 : $R_1 = 26,97\ K\Omega$

$R_2 = 2,128\ K\Omega$

2 : $R_2 = 10,13\ K\Omega = R_2$

$R_1 = 11,17\ K\Omega = R_1$

• Cond $a_2 > 0$; $R_2 + R_1(2-K) = 0 \rightarrow K = 2 + \frac{R_2}{R_1}$

KC_1 : $K_{C_1} = 2,079$

$KC_2 = 3,623$

pour Cellule 1 c'est très proche a l'instabilité

• Refaire : $R_2 = 8,44\ K\Omega$ $K' = 2,313$

$\frac{a_2}{\omega_0} = C(R_2 + R_1(2-K))$

$K = 2 + \frac{R_2}{R_1} - \frac{a_2}{\omega_0 C R_1}$ $K' = 2,988$

I- Etude en Sim :

. Expériment (LTspice):

1) figure 1 : output simulation for butterworth
" 2 : " " Tchébychev

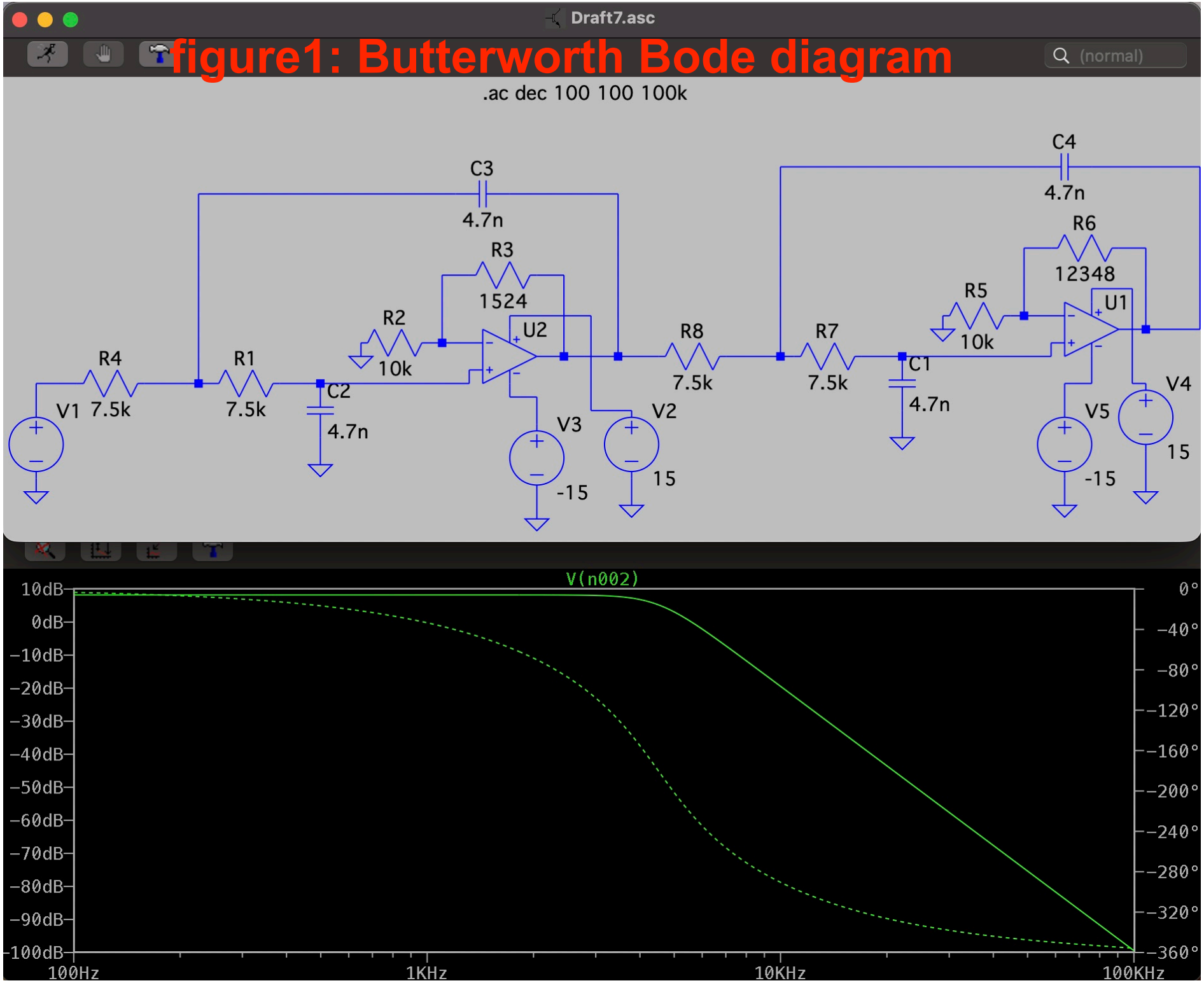
2) from the output figures we can that:
- maximum audintial gain is constant in Butterworth, around 7,5 dB for $f < 3,7$ KHz ; and get to 5db in 4,5 KHz ($\sim$ 3dB)
for tchebychev, gain starts at ~~14,8 dB~~ $\sim$ 14,5 dB, starts oscillation $\pm$ 1dB at 3,5 KH, reaches max at 4,5 KHz with 16,2 dB

→ both start decreasing after, with Tcheby getting to -10 dB at 7 KHz and butterworth at 8 KHz
limits of the ~~eg~~ angels are identical, but Tcheby got a steeper changes

3) . One capacitor changed by 5% : C = 4,5 nF
Tcheby max = 18,1 db, butterworth resists

- R_1 for cellule one changed by 10%, $R_1 = 6f50_{butter}$; $R_1 = f566_{Tcheby}$
Tcheby is less ressistant both in terms of gain and phase ⇒

⇒ butterworth is very resistant to dispersion compared to Tchebychev


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figure1: Butterworth Bode diagram
(normal)
.ac dec 100 100 100k
C3
4.7n
C4
4.7n
R3
1524
R6
12348
R2
10k
U2
R5
10k
U1
R8
7.5k
R7
7.5k
C1
4.7n
R4
7.5k
R1
7.5k
C2
4.7n
V1
V3
-15
V2
15
V5
-15
V4
15
V(n002)
10dB
0dB
-10dB
-20dB
-30dB
-40dB
-50dB
-60dB
-70dB
-80dB
-90dB
-100dB
0°
-40°
-80°
-120°
-160°
-200°
-240°
-280°
-320°
-360°
100Hz
1KHz
10KHz
100KHz

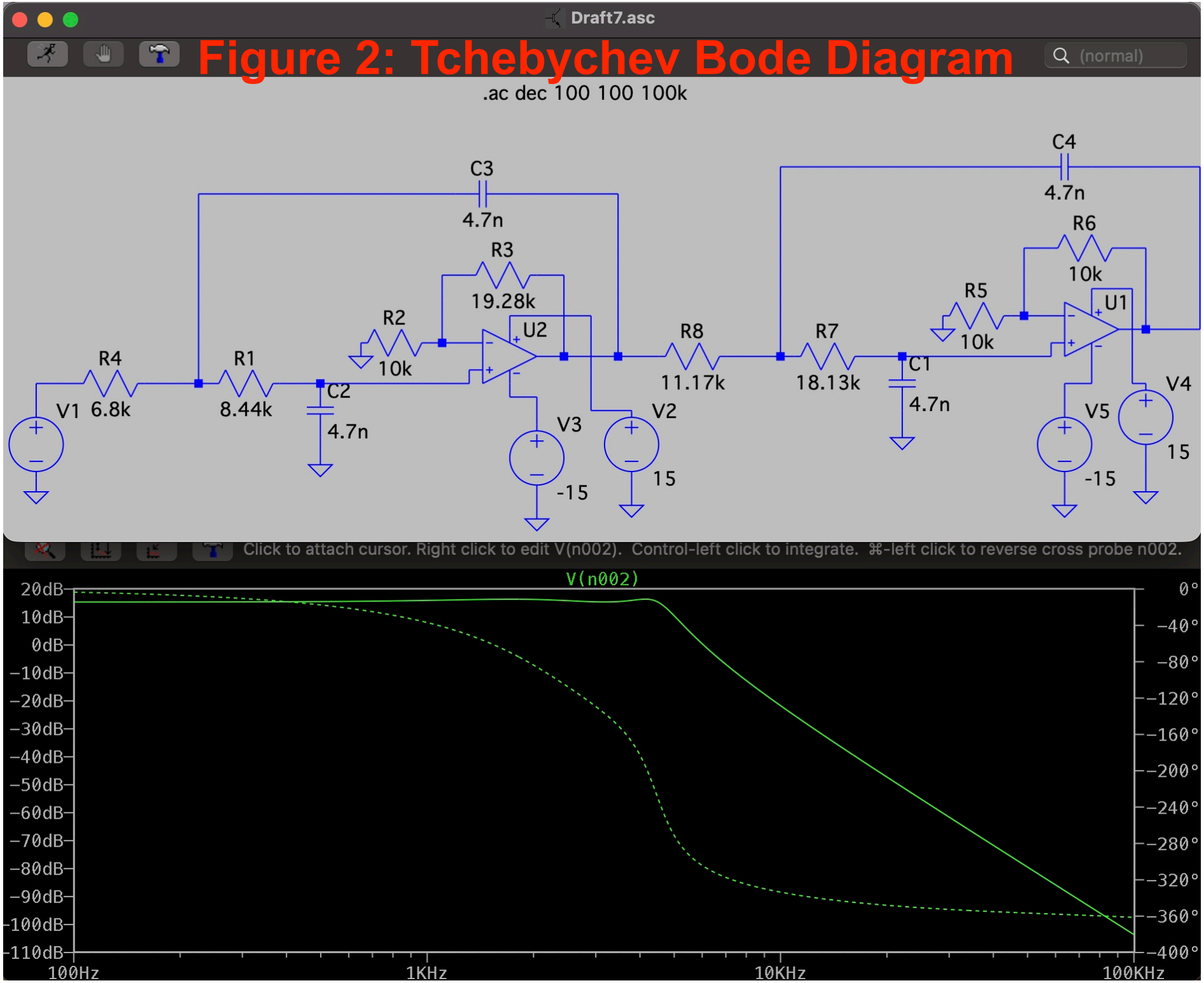

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Figure 2: Tchebychev Bode Diagram
(normal)
.ac dec 100 100 100k
C4
4.7n
C3
4.7n
R6
10k
R3
19.28k
R5
10k
R2
10k
U2
U1
R8
11.17k
R7
18.13k
C1
4.7n
R4
6.8k
R1
8.44k
C2
4.7n
V1
V3
-15
V2
15
V5
-15
V4
15
Click to attach cursor. Right click to edit V(n002). Control-left click to integrate. ⌘-left click to reverse cross probe n002.
V(n002)
20dB
10dB
0dB
-10dB
-20dB
-30dB
-40dB
-50dB
-60dB
-70dB
-80dB
-90dB
-100dB
-110dB
0°
-40°
-80°
-120°
-160°
-200°
-240°
-280°
-320°
-360°
-400°
100Hz
1KHz
10KHz
100KHz

Figure 3: Comparation in dispersion of components

